Eprad eCNA Cinema Automation Plugin

Version 1.2 | Build 49

Use this plugin to monitor and control an Eprad eCNA Cinema Automation system from within your Q-SYS designs. The plugin communicates over two independent channels: the CAI (Cinema Automation Interface) ASCII TCP protocol for commands and status, and a direct HTTP connection for I/O flag detail and, optionally, Eprad QDC-400 dimmer lighting.

What's New in 1.2

Version 1.2 adds optional monitoring and control of Eprad QDC-400 dimmer lighting boards, replicating the eCNA's own *Status: Light Control* web page, plus several changes to existing behavior:

- **QDC-400 Dimmer Lighting** — new optional Dimmer Control page with per-zone preset buttons (Up/Down/Mid1/Mid2/None), an offset fader with dedicated increase/decrease step buttons, and animated per-channel level meters. See QDC-400 Dimmer Lighting.
- **Scrape Interval** — accepts fractional seconds (e.g. 1.6), not just whole seconds. See Properties.
- **Light Poll Interval** — new property controlling how often dimmer status is polled, independently of Scrape Interval. See Light Poll Interval.
- **Faster Digital Projector status** — the four Digital Projector Power/Video buttons now update immediately from the eCNA's own unsolicited event reports rather than waiting on the next scrape. See Digital Projectors.
- **Poll Rate removed** — no longer a configurable property; the plugin now uses a fixed 45-second rate.
- **Auto Refresh Names removed** — input/output names (and, when dimmer lighting is enabled, zone names and QDC board detection) are now fetched once on connect and otherwise only on demand. See Refresh Setup Data.
- **Refresh buttons consolidated** — the separate Refresh Input Names / Refresh Output Names buttons have been replaced by a single Refresh Setup Data button on the Setup page.
- **Debug tab removed** — the Show Debug Tab property, its Debug page, and the Last_Tx/Last_Rx/Last_Error pins are gone; everything they showed is already in the Q-SYS Debug

window. See Show Debug Tab under Properties if you had anything wired to those specific pins.

- **QDC Zones / QDC Channels properties** — now hidden in the Properties dialog until Enable QDC Lighting is turned on, instead of always showing regardless.

Compatibility

This plugin was developed and tested with Q-SYS Designer Software version 9.13.1 LTS, and is intended to be compatible with 9.4.8 LTS and newer. It has been tested against Eprad eCNA firmware v1.23 and v1.26, running on both the eCNA-5 and eCNA-10.

> **Note:** The plugin communicates via the CAI TCP interface (default port 13002) and the eCNA HTTP web interface (port 80). Both must be reachable from the Q-SYS Core on your network. Because the eCNA supports only one CAI TCP connection per port at a time, avoid using the same port number in multiple concurrent designs.

Configuration Overview

Connect the Plugin

1. Drag the Eprad eCNA plugin into your Q-SYS schematic.
2. In the plugin Properties, configure the number of Macros and RDI devices for your installation. If the installation includes Eprad QDC-400 dimmer boards, enable Enable QDC Lighting and set the zone/channel counts — see QDC-400 Dimmer Lighting.
3. Save the design to your Core and run (F5).
4. On the Setup page, enter the IP address of the eCNA and confirm the TCP Port (default 13002).

The plugin connects automatically once a valid IP address is entered. A Connection Status of **OK** confirms a successful connection.

Communication Architecture

The plugin uses two simultaneous communication channels:

- **CAI TCP (port 13002):** The primary control channel. Sends commands (EVT, OUT, RDI, MAC) and receives status responses (RST, RID) and unsolicited event reports (RPT,EVT). A full RST status re-poll is also triggered immediately whenever a relevant event report arrives, and on a fixed internal interval as a background safety net — nearly all status updates arrive via the unsolicited events, not this timer. Only one device may have a port open at a time.
- **HTTP (port 80):** A secondary read-only channel for I/O flag detail, providing the state of all 44 output flags and 31 input flags, including house lights, curtain, masking, lens, sound, and aux outputs. Polled independently at the Scrape Interval; when Enable QDC Lighting is on, this same connection also carries dimmer commands and status at the separate Light Poll Interval.

> **Note:** The HTTP scrape does not consume a CAI port. Multiple Q-SYS designs can scrape the eCNA web interface simultaneously without conflict. Only the CAI TCP connection is exclusive per

> port.

Unsolicited Event Reports

The plugin subscribes to unsolicited event reports from the eCNA (`CFG,RPTON`) on connection. The eCNA sends `RPT,EVT` messages when key state changes occur (START, STOP, CUE, projector power changes, fire stop, etc.). The plugin triggers an immediate RST status re-poll on receipt of these events, providing near-instant UI updates without relying solely on the poll timer.

Properties

Num Macros

Integer, 0–50 (default 50). Sets the number of Macro trigger buttons available on the Macros tab. Set to 0 to hide the Macros tab entirely.

Num RDI

Integer, 0–5 (default 5). Sets the number of RDI (Remote Device Interface) devices. Each device has 16 command buttons. Set to 0 to hide the RDI tab entirely.

Scrape Interval

Decimal, 0.5–20 seconds (default 2). Interval between HTTP requests to the eCNA web interface for I/O flag detail. This is the primary update mechanism for the output and input flag indicators (house lights, curtain, masking, lens, sound, aux outputs, and input flags), and — independently of the Light Poll Interval below — also determines how often that same connection is available to service a queued dimmer command or the periodic dimmer status poll when Enable QDC Lighting is on. Values below 2 seconds may interfere with direct operation of the eCNA device; field testing found values as low as 0.5–1 second to work without issue, if faster flag response is needed.

> **Note:** If you're upgrading from an earlier version, this property was previously called Scrape Rate and only accepted whole seconds. Any custom value you had set will reset to the default (2 seconds) and will need to be re-entered.

Light Poll Interval

Decimal, 0.5–20 seconds (default 3). Interval between HTTP requests to the eCNA for dimmer status (zone offsets, preset state, and channel levels) — see QDC-400 Dimmer Lighting. This property is always shown regardless of the Enable QDC Lighting setting, but only takes effect once that property is turned on. It shares the same HTTP connection as Scrape Interval, but runs on its own independent cadence: I/O flag scraping still happens on every available turn, and a dimmer status poll claims a turn only once its own interval has elapsed, so lowering Scrape Interval no longer indirectly slows dimmer responsiveness the way it once did. Field testing found lowering this to 3 seconds (from the same default as Scrape Interval) noticeably improved dimmer preset/offset responsiveness with no adverse effect; there is no reason to raise it above the default other than temporary testing.

> **Note:** This and Scrape Interval share one HTTP connection, so setting either one very low reduces how often the other effectively runs — there's a practical floor to how fast both can go at

> once.

Show Debug Tab (removed)

> **Removed:** This property, its Debug page, and the underlying Last_Tx / Last_Rx / Last_Error controls have been removed entirely. Everything they ever displayed was already duplicated to the Q-SYS Debug window (Help > Q-SYS Debug) via the plugin's own logging, which remains fully active regardless of this property and is the more capable tool for troubleshooting in any case — it shows a running history rather than just the single most recent value. If you had anything in your design wired to the Last_Tx, Last_Rx, or Last_Error pins specifically (rather than just viewing the Debug tab), that wiring will break on upgrade and will need to be removed or replaced — use the Q-SYS Debug window instead.

Enable QDC Lighting

Boolean (default off). Turns on monitoring and control of Eprad QDC-400 dimmer lighting boards, adding the Dimmer Control page and its associated controls. See QDC-400 Dimmer Lighting for full details.

QDC Zones

Integer, 1–16 (default 2). Number of lighting zones to display on the Dimmer Control page. Zone 1 is always House; Zone 2, if included, is always Stage; any zones above 2 are user-defined on the eCNA. Set to 1 for a House-only installation. Only shown in Properties once Enable QDC Lighting is turned on.

QDC Channels

Integer, 1–16 (default 2). Number of dimmer channel meters to display on the Dimmer Control page. A single QDC-400 board supports 4 channels; more than one board can be daisy-chained. Only shown in Properties once Enable QDC Lighting is turned on.

Controls

Setup Page

IP Address

Text entry. Enter the IP address of the eCNA. The plugin connects automatically when a valid address is entered or changed.

TCP Port

Text entry. The CAI TCP port number (default 13002). Change this to match the CAI channel assigned to this Q-SYS design. Common values: 13000 (channel 1), 13001 (channel 2), 13002 (channel 3), 13003 (channel 4).

> **Note:** Port 13003 (channel 4) requires eCNA firmware version 1.25 or later.

Status LED / Status

Indicator LED and text field, read only. Displays the current connection state:

- **OK** (green): Connected and communicating.
- **Initializing** (blue): Connecting or establishing initial communication.

- **Disconnected / Fault** (red): Connection lost or error received.
- **Missing** (orange): No IP address configured.

Reconnect

Trigger button. Forces an immediate reconnection attempt. Useful if the connection has stalled, the IP address was changed, or the port was changed. Note: changing the TCP Port does not take effect until Reconnect is pressed.

Model

Text field, read only. Displays the eCNA device model returned by the RID command on connection.

Firmware

Text field, read only. Displays the eCNA firmware version returned by the RID command.

Refresh Setup Data

Trigger button. Re-fetches everything that's normally only fetched once, at connect: input flag names, output flag names, and — when Enable QDC Lighting is on — zone names, per-channel dimmer level/fade settings, and the QDC-400 board count. Use this after changing any of that data on the eCNA itself (Setup pages); day-to-day operation does not need it.

> **Note:** This data changes rarely if ever after commissioning, so it is not automatically re-polled in the background.

Outputs Page

The Outputs page contains all eCNA output controls, grouped by type. Toggle buttons both send commands to the eCNA and reflect the current output state, most of them as returned by the web scrape.

Digital Projectors (Dig Proj1 Power, Dig Proj1 Video, Dig Proj2 Power, Dig Proj2 Video)

Toggle buttons. Control and indicate the power and video output state of the two digital projectors. These update as soon as the eCNA reports the change — typically near-instant.

Slide Projector

Toggle button. Controls and indicates the power state of the slide (auxiliary) projector. Updates at the regular Scrape Interval.

House Lights

Mutually exclusive Toggle buttons: None / Down / Mid1 / Mid2 / Up. Sends `OUT HLxxx` commands.

Stage Lights

Mutually exclusive Toggle buttons: None / Down / Up. Sends `OUT SLxxx` commands.

Curtain

Mutually exclusive Toggle buttons: None / Close / Open. Sends `OUT CURTxxx` commands.

Masking

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends `OUT MSKxxx` commands.

Lens

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends `OUT LENxxx` commands.

Sound

Mutually exclusive Toggle buttons: None / Aux1 / Aux2 / Dig1 / Dig2 / Mono / NonSync / SR / SVA. Sends `OUT SNDxxx` commands.

Mute

Independent Toggle button. Sends `OUT MUTEON` or `OUT MUTEOFF`.

Aux Output Flags 1–16

Toggle buttons arranged in two columns of eight. Each row shows the flag number (Out 1–Out 16), the toggle button, and the user-defined name fetched from the eCNA. Sends `OUT OUTnON/OFF` commands.

> **Note:** Names refer to Output Flag names as configured in the eCNA web interface, not physical output names on any I/O board. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Dimmer Control Page

Visible only when Enable QDC Lighting is on. Replicates the eCNA's own *Status: Light Control* web page, with animated level meters. See QDC-400 Dimmer Lighting below for full details on this page's controls and behavior.

Inputs Page

Displays all 31 eCNA input flags as LED indicators. Two columns: General Inputs (In1–In16) on the left and System Inputs on the right. Each General Input row shows the LED, the flag number (In1–In16), and the user-defined name fetched from the eCNA.

> **Note:** Names refer to Input Flag names as configured in the eCNA web interface, not physical input names on any I/O board. Input flags are read-only indicators; they cannot be set from the plugin. To refresh names after changing them on the eCNA, use Refresh Setup Data on the Setup page.

Macros Page

Visible only when Num Macros is greater than 0. Contains trigger buttons for Macros 1 through N, arranged 10 per row. Each press sends `EVT MAC,n` to the eCNA. Text labels appear above each button.

RDI Page

Visible only when Num RDI is greater than 0. Contains trigger buttons for up to 5 RDI (Remote Device Interface) devices, each with 16 commands (labelled RDI 1-1 through RDI 5-16), arranged in two rows of 8 per device. Each press sends `EVT RDI,device,command` to the eCNA. Text labels appear above each button.

Program Page

Transport Controls (Start, Stop, Cue, Abort)

Trigger buttons. Send `EVT STA`, `EVT STP`, `EVT CUE`, and `EVT ABT` commands respectively.

Fault Controls (Set Fault, Clear Fault, Reset Fault, Cancel Alarm)

Trigger buttons. Send `EVT FLT`, `EVT CLR`, `EVT RES`, and `EVT CNL` commands respectively.

Program # / Set Prog / Set & Run

Combo box and trigger buttons. Select a program number (1–9) and press Set Prog to send `EVT PGM,n`, or Set & Run to send `EVT SPR,n` (set and immediately start).

Control State

Text field, read only. Displays the current control state from the RST response: **IDL** (idle) or **RUN** (running).

Stopped State

Text field, read only. Displays the stopped state: **OK**, **STP** (stopped), **FLT** (fault), or **FIR** (fire stop active).

Cue Number

Text field, read only. Displays the current cue number within the running program.

Program Number

Text field, read only. Displays the currently active program number.

Segment Name

Text field, read only. Displays the name of the current program segment.

Segment Index

Text field, read only. Displays the current segment index within the program.

CMD Access

Text field, read only. Displays the CAI command access status: **OK** (full access) or **BLK** (blocked — the eCNA is not accepting CAI commands).

Debug Page (removed)

Removed, along with the Show Debug Tab property that controlled it — see Show Debug Tab under Properties for details.

Input & Output Flag Names

The plugin can display user-defined names for the 16 General Input flags (In1–In16) and the 16 Aux Output flags (Out1–Out16). These names are configured on the eCNA web interface under Setup → Input Flag Names and Setup → Output Flag Names.

How Names Are Fetched

Input and output flag names are retrieved automatically from the eCNA on initial connection, and are not automatically re-fetched afterward — they very rarely change once a system is commissioned. If names are changed on the eCNA later, use Refresh Setup Data on the Setup page to pick up the change.

Manual Refresh

The Refresh Setup Data button on the Setup page immediately refreshes input and output names (along with dimmer zone/channel data, if applicable), typically appearing within a few seconds.

Where Names Appear

- **Inputs page:** Each General Input row (In1–In16) shows the LED indicator, the flag number label, and the user-defined name in a text field alongside it.
- **Outputs page:** Each Aux Output Flag row shows the flag number, the toggle button, and the user-defined name in a text field to the right of the button.
- **Control pins:** `In Names 1–16` and `Out Names 1–16` expose the names as readable output pins for use elsewhere in the Q-SYS design.

> **Note:** Names reflect the flag names as configured in the eCNA, not the physical label on any I/O board terminal. The eCNA supports up to 14 characters per name.

QDC-400 Dimmer Lighting

When Enable QDC Lighting is turned on in Properties, the plugin adds a Dimmer Control page that monitors and controls Eprad QDC-400 dimmer boards, replicating the layout and behavior of the eCNA's own *Status: Light Control* web page.

> **Note:** The CAI TCP protocol has no commands for Zone 3–16 lighting, individual dimmer channel levels, or the light level offset — only House and Stage lighting (already present on the Outputs page in v1.1) are reachable that way. All QDC-400 monitoring and control in this section is done through the eCNA's HTTP interface instead, sharing the same connection used for I/O flag detail.

Zones

Each configured zone (up to QDC Zones, set in Properties) is shown as a row with:

- **Zone name** — House and Stage (zones 1–2) are fixed; zones 3–16 are user-defined on the eCNA (Setup → Zone Names) and fetched automatically.
- **Preset buttons** — five mutually exclusive toggle buttons: Up, Down, Mid1, Mid2, None. Selecting one drives that zone's channels to the level configured for that preset on the eCNA (Setup → QDC-400 Dimmers), and resets the zone's offset to 0.
- **Offset fader and text box** — an adjustment from –100% to +100%, applied on top of the active preset's configured level for every channel assigned to that zone (clamped to 0–100%). When the active preset is None, the offset instead becomes the absolute level directly. Dragging the fader only sends a command to the eCNA once it has been still for about 1 second, to avoid flooding the connection while dragging; the displayed value updates immediately regardless.
- **Offset increase / decrease buttons** — small +/- buttons beside the offset fader and text box. A quick press nudges the offset by 1%; press and hold for about 2 seconds to begin auto-repeating at 1%

every 0.5 second until released, always clamped to –100%/+100%. Like the fader, the displayed value and the channel meter animation both update immediately on every press, but the command actually sent to the eCNA is held until input has been idle for about 1 second — so a quick burst of taps, or a long hold, results in one command for the value you land on, not one per tap.

Dimmer Channels

Each configured channel (up to QDC Channels, set in Properties) is shown as a vertical level meter with:

- **Level meter and percentage** — the channel's current output level, 0–100%.
- **Zone** — the name of the zone this channel is currently assigned to on the eCNA (Setup → QDC-400 Dimmers). More than one channel can be assigned to the same zone.
- **Present indicator** — a small LED that lights once the channel has reported a genuine non-zero level at least once. See Dimmer Presence Detection below.

Animated Level Meters

The eCNA can only be polled for dimmer status every couple of seconds at most (shared with the I/O flag scrape, which is intentionally kept fast and responsive since it is the most frequently used part of the plugin). A real dimmer fade often completes faster than that, so rather than have the meters visibly jump or lag behind, the plugin locally animates each channel toward its expected destination using the actual Level %/Fade-Time settings configured on the eCNA (Setup → QDC-400 Dimmers), the same values the physical dimmer itself uses.

This animation runs whenever a preset change is detected — whether it came from this plugin's own buttons, from the eCNA's own automation (a cinema server sending it a cue, a macro, another control system), or directly at the physical unit. House and Stage changes are typically detected within about a second via the same fast I/O flag scrape used for other outputs; other zones are detected on the next dimmer status poll.

The fade duration is the time to go from whichever preset was previously active to the newly selected one, regardless of how far apart their configured levels are — a short hop and a full 0–100% sweep at the same configured rate take about the same time. Offset adjustments always take about 3 seconds to reach their new level, regardless of which preset is active. Selecting None is effectively instant.

> **Note:** This is a prediction, not a live readout, while a channel is animating — the plugin deliberately keeps showing its own smooth estimate rather than a possibly jumpy raw reading, and simply lets the animation finish on its own schedule. It resyncs with the eCNA's actual reported level as soon as the animation completes.

Dimmer Presence Detection

Not every QDC-400 channel necessarily has a physical dimmer connected and powered — for example during commissioning, or on a system with more channels wired than are actually driving fixtures. If a channel is expected to be at a non-zero level (based on its configured preset and offset) but the eCNA reports 0% two poll cycles in a row, the plugin assumes there's no working dimmer on that channel and stops trying to animate it — showing the honest 0% reading instead of repeatedly predicting a level that never arrives. A single genuine non-zero reading, at any time, immediately clears that assumption and lights the channel's Present indicator.

As a second, independent check, the plugin also scans the eCNA's Local I/O Network (Status → Local I/O Network) for active QDC-400 boards, shown as QDC Boards Found on the Dimmer Control page. If zero active boards are found, every channel is immediately marked as having no dimmer present, without waiting on the per-channel detection above. This scan runs once on connect and otherwise only via Refresh Setup Data — it is not repolled in the background.

Control Pins

The following table lists the control pins available when this plugin is used in a Q-SYS schematic. All pins are optional; connect only those required by your design.

Pin Name	Value	String	Position	Direction
Setup				
IP Address		(text)		Input
TCP Port		(text)		Input
Connection Status	0 1 2 3 4 5	OK (Green) Compromised (Yellow) Fault (Red) Not Present (Grey) Missing (Orange) Initializing (Blue)	—	Output
Status LED	0–5	(same as Status)	—	Output
Connected	0 / 1	Off / On	0 / 1	Output
Reconnect	trigger	—	—	Input
Model		(text)		Output
Firmware Version		(text)		Output
Refresh Setup Data	trigger	—	—	Input
Program State				
Control State		IDL / RUN		Output
Stopped State		OK / STP / FLT / FIR		Output
Cue Number		(text)		Output
Program Number		(text)		Output
Segment Name		(text)		Output
Segment Index		(text)		Output
CAI CMD Access		OK / BLK		Output
Transport Commands				
Cmd Start	trigger	—	—	Input
Cmd Stop	trigger	—	—	Input
Cmd Cue	trigger	—	—	Input
Cmd Abort	trigger	—	—	Input
Cmd Fault	trigger	—	—	Input
Cmd Clear Fault	trigger	—	—	Input

Pin Name	Value	String	Position	Direction
Cmd Reset	trigger	—	—	Input
Cmd Cancel Alarm	trigger	—	—	Input
Set Program	(text / combo box, 1–9)			Input
Cmd Set Program	trigger	—	—	Input
Cmd Set And Run	trigger	—	—	Input
Macros (1 – Num Macros)				
Macros 1 … Macros N	trigger	—	—	Input
RDI (1 – Num RDI × 16)				
RDI 1 … RDI 80	trigger	—	—	Input
Outputs				
House Lights 1–5 (1=None 2=Down 3=Mid1 4=Mid2 5=Up)	0 / 1	Off / On	0 / 1	Input / Output
Stage Lights 1–3 (1=None 2=Down 3=Up)	0 / 1	Off / On	0 / 1	Input / Output
Curtain 1–3 (1=None 2=Close 3=Open)	0 / 1	Off / On	0 / 1	Input / Output
Masking 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Lens 1–4 (1=None 2=Flat 3=Scope 4=Special)	0 / 1	Off / On	0 / 1	Input / Output
Sound 1–9 (1=None 2=Aux1 3=Aux2 4=Dig1 5=Dig2 6=Mono 7=NonSync 8=SR 9=SVA)	0 / 1	Off / On	0 / 1	Input / Output
Mute	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig1 Video	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Power	0 / 1	Off / On	0 / 1	Input / Output
Dig2 Video	0 / 1	Off / On	0 / 1	Input / Output
Aux Projector (Slide Projector)	0 / 1	Off / On	0 / 1	Input / Output
Aux Out 1–16	0 / 1	Off / On	0 / 1	Input / Output
Input & Output Flag Names				
In Names 1–16	(text — user-defined name for In1–In16)			Output
Out Names 1–16	(text — user-defined name for Out1–Out16)			Output
QDC-400 Dimmer Lighting (present only when Enable QDC Lighting is on)				
Zone Name 1 … Zone Name (QDC Zones)	(text — zone name, House/Stage/user-defined)			Output
Zone Preset 1 … Zone Preset (QDC Zones × 5) (5 per zone, in order Up/Down/Mid1/Mid2/None)	0 / 1	Off / On	0 / 1	Input / Output
Zone Offset 1 … Zone Offset (QDC Zones)	-100 – 100	—	—	Input / Output
Zone Offset Text 1 … Zone Offset Text (QDC Zones)	(text, editable, -100 to 100)			Input / Output
Zone Offset Inc 1 … Zone Offset Inc (QDC Zones)	0 / 1	Off / On	0 / 1	Input / Output

Pin Name	Value	String	Position	Direction
Zone Offset Dec 1 … Zone Offset Dec (QDC Zones)	0 / 1	Off / On	0 / 1	Input / Output
Channel Level 1 … Channel Level (QDC Channels)	0 – 100	—	—	Output
Channel Level Text 1 … Channel Level Text (QDC Channels)	(text, e.g. "75%")			Output
Channel Zone 1 … Channel Zone (QDC Channels)	(text — zone name this channel is assigned to)			Output
Channel Present 1 … Channel Present (QDC Channels)	0 / 1	Off / On	0 / 1	Output
QDC Board Count	(text — number of active QDC-400 boards found)			Output

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